

A Practically Stabilizing Model Predictive Control Scheme for Switched Affine Systems

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Abstract—This letter presents the design and analysis of a practically stabilizing model predictive control (MPC) scheme for a class of discrete-time switched affine systems with at least one stable subsystem. A pruning algorithm with cost approximation is proposed to ensure the practical stability of the MPC scheme while reducing the online computations. A criterion for selecting the pruning parameters to ensure the feasibility of the MPC scheme is presented, which is formulated as a convex optimization problem. With the proposed pruning, the value function of the MPC scheme becomes a practical-Lyapunov function which guarantees global practical stability of the switched affine system for arbitrary prediction horizons. The MPC scheme is applied to switching power converters and simulation results are given for a buck converter which is a DC-DC converter.

Index Terms—Model predictive control, practical stability, switched affine system.

I. INTRODUCTION

SWITCHED systems consist of a finite set of subsystems or modes and a switching index that governs switching among modes. Switched affine systems (SASs) is a subclass of switched systems that has many practical applications, especially in power converter design [1], [2]. SASs are characterized by an affine term that makes the equilibrium points of the subsystems to be different and normally it will be different from the desired equilibrium point. In continuous-time SASs asymptotic stabilization at the equilibrium point is possible using an ideal sliding mode with infinite switching frequency [3]. But, in practice, the switching frequency is finite and bounded by some constraints. Consequently, asymptotic stabilization at the equilibrium point is practically impossible unless there exists a subsystem with the equilibrium point the same as the desired equilibrium point. This leads to the idea of practical stabilization in which the objective is to achieve ultimate boundedness of the state trajectory in steady-state [4]. The initial contributions in this area deals with the practical stabilization of continuous-time SASs [4], [5], [6], [7]. Practical stability of continuous-time SASs with bounded

switching frequency is studied in [4], [5], [6] and in [7] dwell time restrictions are used to bound the switching frequency. Recently, practical stabilization of discrete-time SASs is studied by [8], [9], [10] in which an ellipsoidal invariant set of attraction is provided and a switching function is designed to minimize the volume of the set of attraction.

Even though, there are many works on practical stabilization of SASs, stability of MPC schemes for SASs is not investigated much. MPC schemes for switched affine systems [11], switched linear systems [12], and switched nonlinear systems [13] are studied in the past, among which [11] addresses the problem of practical stability. These approaches comes under the category of state-dependent switching since in MPC the switching index is a function of the state vector. Practical stability of MPC for linear time-invariant (LTI) systems with a discrete (quantized) input is investigated in [14], which is a special case of SASs with the same system matrix and different input matrices. This approach is used in [15] to design an MPC scheme with a unit prediction horizon for a subclass of power converters with the same system matrix. Recently, finite control set MPC (FCS-MPC) schemes are used extensively in power converter design [16], [17] which uses an enumeration method to compute the optimal switching sequence. In these approaches, MPC schemes with small prediction horizons are considered, and stability is not investigated.

In this letter, we present an MPC based state-dependent switching scheme for discrete-time SASs and a pruning algorithm is proposed for ensuring practical stability of the MPC scheme. Among the literature, our work is more related to [8], [9], [10], [14], since we follow a discrete-time SAS framework as in [8], [9], [10], and a practical-Lyapunov function approach is used to prove stability as in [14]. Compared to this literature the contributions in this letter are

- 1) To ensure practical stability of the MPC, a pruning approach with cost approximation is proposed which is performed offline. In the approach, SASs with at least one stable subsystem is considered for which the system matrix can be different for each subsystem.
- 2) Sufficient conditions for ensuring the feasibility of the pruning approach is proposed which is in the form of a convex optimization problem with linear matrix inequality (LMI) constraints and can be solved easily. Moreover, with the proposed pruning approach the value function of the MPC becomes a practical-Lyapunov function, resulting in global practical stability of the SAS for all prediction horizons.

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Notations: $\mathbb{N}, \mathbb{Z}^+, \mathbb{R}^+$ represents the set of natural numbers, nonnegative integers and real numbers. $\mathbb{R}^n$ denotes n - dimensional Euclidean space and $\mathbb{R}^{m \times n}$ refers to the space of $m \times n$ real matrices. $[\mathbf{s}]_i$ denotes the i^{th} element of vector $\mathbf{s}$, and $[\mathbb{S}]_i$ represents the i^{th} element of set $\mathbb{S}$ with the elements ordered by an index. $|\mathbb{S}|$ denotes the cardinality of the set $\mathbb{S}$ and $\|\mathbf{s}\|_p$ denotes the p - norm. $\mathbf{P} > 0$ ($\mathbf{P} \geq 0$) means that $\mathbf{P}$ is a real symmetric positive definite (semidefinite) matrix. The system matrix $\mathbf{A}$ is said to be Schur stable if $\rho(\mathbf{A}) < 1$, where $\rho(\mathbf{A})$ is the spectral radius of $\mathbf{A}$. A function $\alpha : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is called a K - function if it is strictly increasing continuous function with $\alpha(0) = 0$. Similarly, α is a K_∞ function if it is an unbounded K - function, i.e., $\lim_{t \rightarrow \infty} \alpha(t) = \infty$. Finally, $\beta : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is a KL - function if $\beta(x, t)$ is a K - function in x for each fixed t and $\beta(x, t)$ is a non-increasing function in t for each fixed x which satisfies $\lim_{t \rightarrow \infty} \beta(x, t) = 0$.

II. PROBLEM FORMULATION

Consider the discrete-time SAS

$$\mathbf{x}_{k+1} = \mathbf{A}_{\sigma_k} \mathbf{x}_k + \mathbf{B}_{\sigma_k} \quad (1)$$

where $k \in \mathbb{Z}^+$ is the discrete time instant, $\mathbf{x}_k \in \mathbb{R}^n$ is the state vector, $\sigma_k : \mathbb{Z}^+ \rightarrow \mathbb{M} = \{1, 2, \dots, m\}$ is the switching index and $\mathbf{A}_v, \mathbf{B}_v$, are the system matrix and input matrix for subsystem $v \in \mathbb{M}$. The switching sequence is denoted by $\pi = \{\sigma_k\}_{k=0}^\infty$ and if σ_k is a function of $\mathbf{x}_k$ then π is called a switching policy. By defining the augmented state $\bar{\mathbf{x}} = \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} \in \mathbb{R}^{n+1}$, the state equation (1) is compactly represented as

$$\bar{\mathbf{x}}_{k+1} = \bar{\mathbf{A}}_{\sigma_k} \bar{\mathbf{x}}_k \quad (2)$$

where $\bar{\mathbf{A}}_v = \begin{bmatrix} \mathbf{A}_v & \mathbf{B}_v \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}$ is the augmented system matrix for subsystem $v \in \mathbb{M}$.

In MPC, we use a prediction horizon $N \in \mathbb{N}$ (usually $N \geq 2$), and during each time instant, a switching sequence of length N is computed, which minimizes the cost over the prediction horizon. Then a receding horizon scheme with state-dependent switching is constructed by applying the first element of the switching sequence to the system. We define the switching sequence at time instant k for the MPC as $\sigma_k = (\sigma_{k|k}, \sigma_{k+1|k}, \dots, \sigma_{k+N-1|k})$ in which $\sigma_{l|k}, l = k, k+1, \dots, k+N-1$ is the switching index at l^{th} instant computed/predicted at k^{th} instant. Similarly, the cost function at time instant k for the MPC is defined as

$$J_k = \sum_{l=k}^{k+N-1} \mathbf{x}_{l|k}^T \mathbf{Q}_{\sigma_{l|k}} \mathbf{x}_{l|k} = \sum_{l=k}^{k+N-1} \bar{\mathbf{x}}_{l|k}^T \bar{\mathbf{Q}}_{\sigma_{l|k}} \bar{\mathbf{x}}_{l|k} \quad (3)$$

where $\bar{\mathbf{Q}}_v = \begin{bmatrix} \mathbf{Q}_v & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}$ is the augmented weighting matrix for subsystem $v \in \mathbb{M}$. Then the MPC problem for SASs is defined as

Problem 1: For the SAS (1) with the current state $\mathbf{x}_{k|k} = \mathbf{x}_k$ given, compute σ_k by solving:

$$\begin{aligned} \min_{\sigma_k} \quad & J_k \text{ subject to} \\ & \sigma_k \in \mathbb{M}^N \\ & \mathbf{x}_{l+1|k} = \mathbf{A}_{\sigma_{l|k}} \mathbf{x}_{l|k} + \mathbf{B}_{\sigma_{l|k}} \end{aligned} \quad (4)$$

where $k \in \mathbb{Z}^+$, $\sigma_{l|k} = [\sigma_k]_{l-k+1}$ and $l = k, k+1, \dots, k+N-1$. Here σ_k comes from a finite set $\mathbb{M}^N$ containing m^N switching sequences, which makes Problem 1 a finite-control set MPC (FCS-MPC) problem.

III. PRELIMINARIES

A. Practical Stability of Discrete-Time SAS

Without any loss of generality, consider the desired equilibrium point/operating point $\mathbf{x}_o = \mathbf{0}$. In general, any non-zero $\mathbf{x}_o$ in $\mathbb{X}_e$ (where $\mathbb{X}_e$ is the set of attainable equilibrium points as defined in [8]) can be shifted to origin using a coordinate transformation. In the case of discrete-time SASs, the equilibrium point of each of the subsystems is given by $\mathbf{x}_{o_v} = (\mathbf{I}_n - \mathbf{A}_v)^{-1} \mathbf{B}_v$ and it will be different from the origin if $\mathbf{B}_v \neq \mathbf{0}$. In this letter we consider $\mathbf{B}_v \neq \mathbf{0}$, for all $v \in \mathbb{M}$. Consequently, asymptotic stabilization of (1) at origin can not be achieved and in steady-state we have to switch between the subsystems to stay near the operating point $\mathbf{x}_o = \mathbf{0}$. This leads to the idea of practical stability, and from [14] we have the following definitions and a theorem on practical stability.

Definition 1: The SAS (1) under a switching policy π is said to be uniform practical asymptotic stable (UpAS) in $\mathbb{X} \subseteq \mathbb{R}^n$, if $\mathbb{X}$ is a positive invariant set for (1) and there exists a nonnegative constant $\delta \geq 0$ and a KL - function β such that the solution $\mathbf{x}_k = \mathbf{x}(k, \mathbf{x}_0, \pi)$ satisfies

$$\|\mathbf{x}_k\|_2 \leq \beta(\|\mathbf{x}_0\|_2, k) + \delta, \quad \forall \mathbf{x}_0 \in \mathbb{X}, k \in \mathbb{Z}^+. \quad (5)$$

If $\mathbb{X} = \mathbb{R}^n$, then the SAS is said to be globally UpAS.

Definition 2: A function $V : \mathbb{R}^n \rightarrow \mathbb{R}^+$ is called a practical-Lyapunov function in $\mathbb{X}$ for the discrete-time SAS (1) under a switching policy π , if $\mathbb{X}$ is a positive invariant set and there exists a compact set $\mathbb{V} \subseteq \mathbb{X}$ which contains the equilibrium point $\mathbf{x}_o = \mathbf{0}$, and some constants $c_1 \geq 0, c_2 \geq 0, K_\infty$ functions $\alpha_1, \alpha_2, \alpha_3$ such that

$$\begin{aligned} V(\|\mathbf{x}_k\|_2) &\geq \alpha_1(\|\mathbf{x}_k\|_2), \quad \forall \mathbf{x}_k \in \mathbb{X} \\ V(\|\mathbf{x}_k\|_2) &\leq \alpha_2(\|\mathbf{x}_k\|_2) + c_1, \quad \forall \mathbf{x}_k \in \mathbb{V} \\ \Delta V &= V(\mathbf{x}_{k+1}) - V(\mathbf{x}_k) \leq -\alpha_3(\|\mathbf{x}_k\|_2) + c_2, \quad \forall \mathbf{x}_k \in \mathbb{X}. \end{aligned} \quad (6)$$

Theorem 1: If the SAS under a switching policy π admits a practical-Lyapunov function in $\mathbb{X}$, then it is UpAS in $\mathbb{X}$. If $\mathbb{X} = \mathbb{R}^n$, then the SAS is said to be globally UpAS.

B. Enumeration Based MPC

Enumeration-based MPC (also known as exhaustive search-based MPC) is an approach used to solve the FCS-MPC problem and is extensively used in power converter design [16], [17]. The approach is based on an exhaustive search among all the switching sequences to compute the optimal sequence. Here a switching tree of length N as in Fig. 1 is considered which consists of m^N switching sequences, viz: $\mathbf{s}_1 = \{1, 1, \dots, 1\}$ to $\mathbf{s}_{m^N} = \{m, m, \dots, m\}$. For a fixed switching sequence, system (2) becomes a deterministic linear time-varying (LTV) system. Consequently, we have the following Lemma which is a special case of [18, Lemma 1] with control input and feedback gain zero, hence the proof is omitted.

Lemma 1: For the SAS with a given switching sequence σ_k , the cost function is given by

$$J_k(\sigma_k) = \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_{k|k} \bar{\mathbf{x}}_k \quad (7)$$

in which the matrix $\bar{\mathbf{P}}_{k|k}$ is computed recursively as

$$\bar{\mathbf{P}}_{l|k} = \bar{\mathbf{Q}}_{\sigma_{l|k}} + \bar{\mathbf{A}}_{\sigma_{l|k}}^T \bar{\mathbf{P}}_{l+1|k} \bar{\mathbf{A}}_{\sigma_{l|k}} \quad (8)$$

where $l = k + N - 1, \dots, k + 1, k$, $\bar{\mathbf{P}}_{k+N|k} = \mathbf{0}_{(n+1) \times (n+1)}$ and $\sigma_{l|k} = [\sigma_k]_{l-k+1}$. And the optimal cost is

$$J_k^* = \min_{\sigma_k \in \mathbb{M}^N} J_k(\sigma_k) = \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_{k|k}(\sigma_k^*) \bar{\mathbf{x}}_k. \quad (9)$$

where $\sigma_k^* = \operatorname{argmin}_{\sigma_k \in \mathbb{M}^N} J_k(\sigma_k)$.

Let $\bar{\mathbf{P}}_{l|k} = \begin{bmatrix} \mathbf{P}_{l|k} & \mathbf{h}_{l|k} \\ \mathbf{h}_{l|k}^T & d_{l|k} \end{bmatrix}$, then by expanding and equating the terms in (8) we obtain

$$\begin{aligned} \mathbf{P}_{l|k} &= \mathbf{Q}_{\sigma_{l|k}} + \mathbf{A}_{\sigma_{l|k}}^T \mathbf{P}_{l+1|k} \mathbf{A}_{\sigma_{l|k}} \\ \mathbf{h}_{l|k} &= \mathbf{A}_{\sigma_{l|k}}^T \mathbf{P}_{l+1|k} \mathbf{B}_{\sigma_{l|k}} + \mathbf{A}_{\sigma_{l|k}}^T \mathbf{h}_{l+1|k} \\ d_{l|k} &= \mathbf{B}_{\sigma_{l|k}}^T \mathbf{P}_{l+1|k} \mathbf{B}_{\sigma_{l|k}} + 2\mathbf{B}_{\sigma_{l|k}}^T \mathbf{h}_{l+1|k} + d_{l+1|k} \end{aligned} \quad (10)$$

where $\mathbf{P}_{k+N|k} = \mathbf{0}_{n \times n}$, $\mathbf{h}_{k+N|k} = \mathbf{0}_{n \times 1}$ and $d_{k+N|k} = 0$.

In the case of the MPC scheme with a fixed prediction horizon N , we can use the same switching tree for $k = 0$ as in Fig. 1 which shifts along with the time instant k and contains m^N switching sequences. To simplify the notation we set $s_j = [s_j]_1$, $\bar{\mathbf{P}}_j = \bar{\mathbf{P}}_{0,j}$, $\mathbf{A}_j = \mathbf{A}_{s_j}$, $\mathbf{B}_j = \mathbf{B}_{s_j}$, $j = 1, 2, \dots, m^N$. Define $\sigma = \{s_1, s_2, \dots, s_{m^N}\}$, $\mathbb{P} = \{\bar{\mathbf{P}}_1, \bar{\mathbf{P}}_2, \dots, \bar{\mathbf{P}}_{m^N}\}$. The next lemma is directly obtained from [19, Lemma 4.22].

Lemma 2: For the MPC scheme with a fixed terminal weighting matrix and prediction horizon, the matrix $\bar{\mathbf{P}}_{k|k} \in \mathbb{P}$ for all $k \in \mathbb{Z}^+$. Consequently, the optimal cost J_k^* becomes

$$J_k^* = \min_{j=1,2,\dots,m^N} \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_j \bar{\mathbf{x}}_k, \quad \bar{\mathbf{P}}_j \in \mathbb{P}. \quad (11)$$

Therefore we can implement the enumeration-based MPC in the following two ways.

- 1) In this approach, the cost for each switching sequence is computed online using (3) and the cost is compared to compute the minimum cost. The corresponding switching sequence will be the optimal one.
- 2) In the second approach, the sets $\sigma, \mathbb{P}$ are computed and stored offline using (8). Then during the online computation, the minimum cost is computed using (11).

In MPC, one of the major approaches for ensuring stability is to consider the value function J_k^* as a candidate Lyapunov function. For enumeration based MPC, we have $J_k^* = \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_k$, $j_k = \operatorname{argmin}_{j=1,2,\dots,m^N} \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_j \bar{\mathbf{x}}_k$. This gives the candidate Lyapunov function $V(\mathbf{x}_k) = \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_k$ and

$$\begin{aligned} \Delta V &= \bar{\mathbf{x}}_{k+1}^T \bar{\mathbf{P}}_{j_{k+1}} \bar{\mathbf{x}}_{k+1} - \bar{\mathbf{x}}_k^T \bar{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_k \\ &= \begin{bmatrix} \mathbf{A}_{j_k} \mathbf{x}_k + \mathbf{B}_{j_k} \\ 1 \end{bmatrix}^T \begin{bmatrix} \mathbf{P}_{j_{k+1}} & \mathbf{h}_{j_{k+1}} \\ \mathbf{h}_{j_{k+1}}^T & d_{j_{k+1}} \end{bmatrix} \begin{bmatrix} \mathbf{A}_{j_k} \mathbf{x}_k + \mathbf{B}_{j_k} \\ 1 \end{bmatrix} \\ &\quad - \begin{bmatrix} \mathbf{x}_k \\ 1 \end{bmatrix}^T \begin{bmatrix} \mathbf{P}_{j_k} & \mathbf{h}_{j_k} \\ \mathbf{h}_{j_k}^T & d_{j_k} \end{bmatrix} \begin{bmatrix} \mathbf{x}_k \\ 1 \end{bmatrix} \\ &= \mathbf{x}_k^T [\mathbf{A}_{j_k}^T \mathbf{P}_{j_{k+1}} \mathbf{A}_{j_k} - \mathbf{P}_{j_k}] \mathbf{x}_k \\ &\quad + 2\mathbf{x}_k^T [\mathbf{A}_{j_k}^T \mathbf{P}_{j_{k+1}} \mathbf{B}_{j_k} + \mathbf{A}_{j_k}^T \mathbf{h}_{j_{k+1}} - \mathbf{h}_{j_k}] \\ &\quad + \mathbf{B}_{j_k}^T \mathbf{P}_{j_{k+1}} \mathbf{B}_{j_k} + 2\mathbf{B}_{j_k}^T \mathbf{h}_{j_{k+1}} + d_{j_{k+1}} - d_{j_k}. \end{aligned} \quad (12)$$

In this case, bounding ΔV as in (6) is difficult because of the linear term in $\mathbf{x}_k$. Consequently, constructing a practical

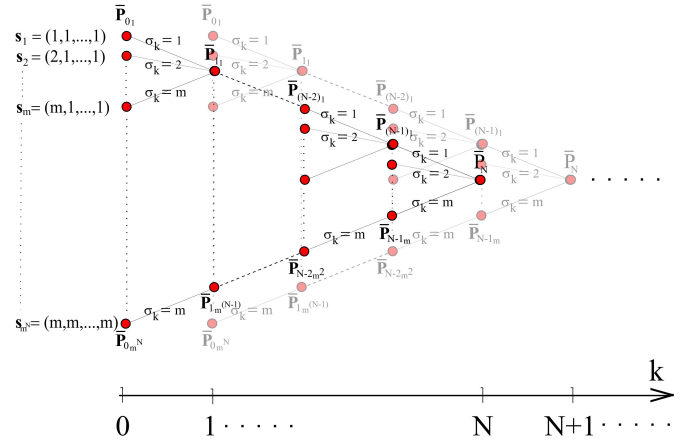


Fig. 1. Switching tree for MPC of SASs with m subsystems.

Lyapunov function for the enumeration-based MPC scheme is difficult. Therefore, in this letter, we use a pruning approach with cost approximation to bound ΔV which makes the linear term in $\mathbf{x}_k$ zero, and the value function of the MPC scheme results in a practical Lyapunov function.

IV. PROPOSED MPC SCHEME

The proposed MPC scheme consists of an offline computation and an online computation part. We define $\mathbb{M}_s$ as the set of stable subsystems, i.e., $\mathbb{M}_s = \{v \in \mathbb{M} : \mathbf{A}_v \text{ is Schur stable}\}$. In order to guarantee feasibility of the MPC scheme, the following assumption is used:

Assumption 1: $\mathbb{M}_s$ is non-empty.

A. Offline Computation

To ensure practical stability of the proposed MPC scheme, we introduce the set $\sigma_P \subseteq \sigma$ as the feasible set of switching indices and the corresponding approximated initial matrices are stored in $\mathbb{P}_P$. The sets $\sigma_P, \mathbb{P}_P$ are computed offline using the pruning algorithm which is summarized as below:

- 1) Select the first subsystem from the set $\mathbb{M}_s$ and choose $\mathbf{W} > 0, C > 0$ such that

$$\begin{aligned} \mathbf{A}_1^T \mathbf{P}_1 \mathbf{A}_1 - \mathbf{P}_1 &\leq -\mathbf{W}, \hat{\mathbf{P}}_1 > 0, \& \\ \mathbf{B}_1^T \mathbf{P}_1 \mathbf{B}_1 + \mathbf{B}_1^T \hat{\mathbf{h}}_1 &\leq C \end{aligned} \quad (13)$$

where $\hat{\mathbf{P}}_1 = \begin{bmatrix} \mathbf{P}_1 & \hat{\mathbf{h}}_1 \\ \hat{\mathbf{h}}_1^T & d_1 \end{bmatrix}$ is the approximate initial matrix and

$$\hat{\mathbf{h}}_1 = (\mathbf{I}_n - \mathbf{A}_1^T)^{-1} \mathbf{A}_1^T \mathbf{P}_1 \mathbf{B}_1. \quad (14)$$

Initialize $\sigma_P = \{s_1\}$, $\mathbb{P}_P = \{\hat{\mathbf{P}}_1\}$. The selection of $\mathbf{W}, C$ will be discussed in Section V.

- 2) Then for $j = 2, \dots, m^N$, if there exists a $\hat{\mathbf{P}}_j \in \mathbb{P}_P$ satisfying

$$\begin{aligned} \mathbf{A}_j^T \mathbf{P}_j \mathbf{A}_j - \mathbf{P}_j &\leq -\mathbf{W}, \hat{\mathbf{P}}_j > 0, \& \\ \mathbf{B}_j^T \mathbf{P}_j \mathbf{B}_j + \mathbf{B}_j^T \hat{\mathbf{h}}_j + d_j - d_j &\leq C \end{aligned} \quad (15)$$

where $\hat{\mathbf{P}}_j = \begin{bmatrix} \mathbf{P}_j & \hat{\mathbf{h}}_j \\ \hat{\mathbf{h}}_j^T & d_j \end{bmatrix}$ is the approximate initial matrix and

$$\hat{\mathbf{h}}_j = \mathbf{A}_j^T \mathbf{P}_i \mathbf{B}_j + \mathbf{A}_j^T \hat{\mathbf{h}}_i \quad (16)$$

then update $\mathcal{O}_P \leftarrow \mathcal{O}_P \cup \{s_j\}$, $\mathbb{P}_P \leftarrow \mathbb{P}_P \cup \{\hat{\mathbf{P}}_j\}$.

Note that (14) comes from (16) by substituting $j = 1, i = 1$ and solving for $\hat{\mathbf{h}}_1$. The approximation $\hat{\mathbf{h}}_1$ is similar to the selection of $\mathbf{h}_*$ in [8] whereas here we extend the idea in [8] to the receding horizon switching problem. Once $\mathbf{W}$ and C are selected, we prune the matrices by checking the inequalities in (15) in which the first inequality can be rewritten as

$$\mathbf{P}_j - \mathbf{A}_j^T \mathbf{P}_i \mathbf{A}_j - \mathbf{W} \geq 0. \quad (17)$$

This and the second inequality in (15) are matrix inequalities that can be checked using any numerical conditions for checking positive definiteness.

Remark 1: The approximation $\hat{\mathbf{h}}_j$ in (16) is of the same form as $\mathbf{h}_j$ in (10) with same $\mathbf{A}_j, \mathbf{B}_j$. An upper bound for the approximated cost is obtained easily as below

$$\begin{aligned} \bar{\mathbf{x}}_k^T \hat{\mathbf{P}}_j \bar{\mathbf{x}}_k &= \bar{\mathbf{x}}_k^T \mathbf{P}_j \bar{\mathbf{x}}_k + 2\bar{\mathbf{x}}_k^T \hat{\mathbf{h}}_j + d_j \\ &= \bar{\mathbf{x}}_k^T \mathbf{P}_j \bar{\mathbf{x}}_k + 2\bar{\mathbf{x}}_k^T \mathbf{h}_j + d_j + 2\bar{\mathbf{x}}_k^T \hat{\mathbf{h}}_j - 2\bar{\mathbf{x}}_k^T \mathbf{h}_j \\ &\leq \bar{\mathbf{x}}_k^T \hat{\mathbf{P}}_j \bar{\mathbf{x}}_k + d \|\bar{\mathbf{x}}_k\| \end{aligned} \quad (18)$$

where $d = 2\|\hat{\mathbf{h}}_j - \mathbf{h}_j\|_\infty$ which can be computed offline.

B. Online Computation

During the online computation, we use the feasible set of switching indices $\mathcal{O}_P$ and the corresponding set of approximated initial matrices $\mathbb{P}_P$ for computing the switching scheme. Let $\mu_P = |\mathcal{O}_P| = |\mathbb{P}_P|$ and $\mathbb{W} = \{1, 2, \dots, \mu_P\}$. We assume that all the states are available for feedback. Then the switching index for the MPC scheme is computed at each time instant using the current state $\mathbf{x}_{k|k} = \mathbf{x}_k$ as follows:

$$\begin{aligned} j_k &= \underset{j \in \mathbb{W}}{\operatorname{argmin}} \bar{\mathbf{x}}_k^T \hat{\mathbf{P}}_j \bar{\mathbf{x}}_k, \quad \hat{\mathbf{P}}_j \in \mathbb{P}_P \\ \sigma_{k|k} &= [\mathcal{O}_P]_{j_k}. \end{aligned} \quad (19)$$

We denote $\pi_{MPC} = \{\sigma_{k|k}\}_{k=0}^\infty$ as the receding horizon switching policy computed using (19).

Remark 2: The MPC scheme results in state-dependent switching, since the computation of $\sigma_{k|k}$ in (19) requires the current state information. The online computation and storage required for the MPC scheme is of order μ_P . Moreover, the MPC scheme permits switching to unstable subsystems, since the inequality (17) can be satisfied by an unstable $\mathbf{A}_j$ if there exists a $\mathbf{P}_i$ that is sufficiently small compared to $\mathbf{P}_j$.

V. ANALYSIS OF THE MPC SCHEME

A. Feasibility Analysis

To ensure the feasibility of the pruning and MPC scheme one has to ensure that the sets $\mathcal{O}_P, \mathbb{P}_P$ are nonempty. Under Assumption 1, we can ensure the feasibility of the pruning and the MPC scheme by properly selecting the pruning parameters $\mathbf{W}, C$, and the weighting matrix $\mathbf{Q}_1$ corresponds to subsystem 1. This is given in the next theorem.

Theorem 2: Consider the SAS (1) and let $\mathbf{A}_1, \mathbf{B}_1$ be a Schur stable subsystem. For a given $N \in \mathbb{N}$, choose $C > 0$ and compute $\mathbf{W}, \mathbf{Q}_1$ as solutions to the optimization problem:

$$\begin{aligned} \min_{\mathbf{Q}_1} \quad & -\lambda_{\min}(\mathbf{W}) \text{ subject to} \\ & \mathbf{Q}_1 - \mathbf{A}_1^{N^T} \mathbf{Q}_1 \mathbf{A}_1^N - \mathbf{W} \geq 0 \\ & \sum_{z=0}^{N-1} \hat{\mathbf{B}}_1^T \mathbf{A}_1^z \mathbf{Q}_1 \mathbf{A}_1^z \mathbf{B}_1 \leq C, \mathbf{W} > 0, \mathbf{Q}_1 > 0 \end{aligned} \quad (20)$$

where $\hat{\mathbf{B}}_1^T = \mathbf{B}_1^T (\mathbf{I}_n + 2(\mathbf{I}_n - \mathbf{A}_1^T)^{-1} \mathbf{A}_1^T)$. Then the pruning algorithm with the parameters $\mathbf{W}, C, \mathbf{Q}_1$ is feasible. Consequently, the MPC scheme is feasible.

Proof: To prove the theorem we consider the switching sequence $(1, 1, \dots, 1) \in \mathbb{M}^N$ and show that the corresponding initial matrix will be an element of $\mathbb{P}_P$. For $(1, 1, \dots, 1)$ using (10), (14) we obtain

$$\begin{aligned} \mathbf{P}_1 &= \mathbf{Q}_1 + \mathbf{A}_1^T \mathbf{Q}_1 \mathbf{A}_1 + \dots + \mathbf{A}_1^{N-1^T} \mathbf{Q}_1 \mathbf{A}_1^{N-1} \\ \mathbf{B}_1^T \mathbf{P}_1 \mathbf{B}_1 + \mathbf{B}_1^T \hat{\mathbf{h}}_1 &= \hat{\mathbf{B}}_1^T \mathbf{P}_1 \mathbf{B}_1 \end{aligned} \quad (21)$$

which leads to

$$\begin{aligned} \mathbf{A}_1^T \mathbf{P}_1 \mathbf{A}_1 - \mathbf{P}_1 &= \mathbf{A}_1^{N^T} \mathbf{Q}_1 \mathbf{A}_1^N - \mathbf{Q}_1 \\ \mathbf{B}_1^T \mathbf{P}_1 \mathbf{B}_1 + \mathbf{B}_1^T \hat{\mathbf{h}}_1 &= \sum_{z=0}^{N-1} \hat{\mathbf{B}}_1^T \mathbf{A}_1^z \mathbf{Q}_1 \mathbf{A}_1^z \mathbf{B}_1. \end{aligned} \quad (22)$$

Now, if we select $\mathbf{W}, C$ and $\mathbf{Q}_1$ as solutions to (20), then the initial matrix for $(1, 1, \dots, 1)$ satisfies (13) and the corresponding approximated matrix will be an element of $\mathbb{P}_P$. Note that (20) is a convex optimization problem with LMI constraints, hence a solution always exists for a stable $\mathbf{A}_1$ [8], which ensures the feasibility of the MPC scheme for all $N \in \mathbb{N}$. ■

B. Stability analysis

In MPC of SASs proving stability is difficult because of the following reasons: 1. Receding horizon nature of the MPC, 2. Time-varying and affine dynamics of the SAS. In this letter, we used a pruning algorithm which results in a practically stabilizing MPC scheme. This leads to the next theorem.

Theorem 3: Under Assumption 1, the MPC scheme π_{MPC} makes the SAS globally UpAS for any $N \in \mathbb{N}$.

Proof: To prove this theorem, it is sufficient to find a practical-Lyapunov function for the MPC scheme π_{MPC} for $N \in \mathbb{N}$. We consider the value function for the MPC scheme as a candidate Lyapunov function and from (19) we have

$$V(\mathbf{x}_k) = \bar{\mathbf{x}}_k^T \hat{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_k = \begin{bmatrix} \mathbf{x}_k \\ 1 \end{bmatrix}^T \begin{bmatrix} \mathbf{P}_{j_k} & \hat{\mathbf{h}}_{j_k} \\ \hat{\mathbf{h}}_{j_k}^T & d_{j_k} \end{bmatrix} \begin{bmatrix} \mathbf{x}_k \\ 1 \end{bmatrix}. \quad (23)$$

We have $\lambda_{\min}(\mathbf{P}) \|\mathbf{x}\|_2^2 \leq \mathbf{x}^T \mathbf{P} \mathbf{x} \leq \lambda_{\max}(\mathbf{P}) \|\mathbf{x}\|_2^2$, for any $\mathbf{P} > 0$ [20]. This result in

$$a_1 \|\bar{\mathbf{x}}_k\|_2^2 \leq V(\mathbf{x}_k) \leq a_2 \|\bar{\mathbf{x}}_k\|_2^2 \quad (24)$$

where $a_1 = \min_{j \in \mathbb{W}} \lambda_{\min}(\hat{\mathbf{P}}_j)$ and $a_2 = \max_{j \in \mathbb{W}} \lambda_{\max}(\hat{\mathbf{P}}_j)$, $\hat{\mathbf{P}}_j \in \mathbb{P}_P$. From the definition of the augmented state vector we obtain $\|\bar{\mathbf{x}}_k\|_2^2 = \|\mathbf{x}_k\|_2^2 + 1$. Substituting this in (24) gives

$$V(\mathbf{x}_k) \geq a_1 \|\mathbf{x}_k\|_2^2 + a_1 \geq a_1 \|\mathbf{x}_k\|_2^2 = \alpha_1 (\|\mathbf{x}_k\|_2)$$

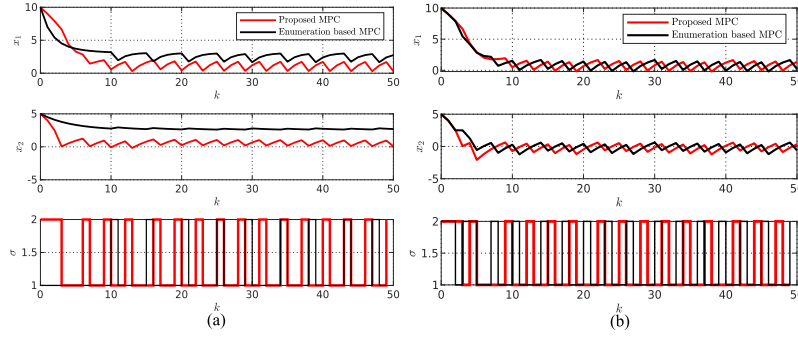


Fig. 2. Response of the SAS (27) with proposed MPC and enumeration-based MPC for (a) $N = 5$ (b) $N = 10$.

TABLE I
PERFORMANCE COMPARISON OF THE DISCRETE-TIME SAS (27)

(a) Proposed MPC								(b) Enumeration-based MPC						
N	$\mathbf{W}$		$\mathbf{Q}_1$		μ_P	t_P	RC	J	N	$\mathbf{Q}_1$		μ_E	t_E	J
2	0.2060	0	0.3138	0	2	0.0059	1.6	408.7	2	0.3138	0	4	0.0060	447.8
	0	0.2060	0	0.5328						0	0.5328			
3	0.2063	0	0.2876	0	6	0.0066	8.3	138.0	3	0.2876	0	8	0.0072	348.1
	0	0.2063	0	0.4091						0	0.4091			
5	0.2064	0	0.2758	0	15	0.0073	22.3	127.1	5	0.2758	0	32	0.0094	251.6
	0	0.2064	0	0.3206						0	0.3206			
10	0.3265	0	0.3276	0	806	0.0790	17.4	144.6	10	0.3276	0	1024	0.0956	140.7
	0	0.3265	0	0.3331						0	0.3331			

$$V(\mathbf{x}_k) \leq a_2 \|\mathbf{x}_k\|_2^2 + a_2 = \alpha_2 (\|\mathbf{x}_k\|_2) + c_1. \quad (25)$$

We denote $\mathbf{A}_{\sigma_{k|k}} = \mathbf{A}_{[\sigma_P]_{j_k}} = \mathbf{A}_{j_k}$ to simplify the notation. Then from (19), (15), and (16), we have

$$\begin{aligned} \Delta V &= \bar{\mathbf{x}}_{k+1}^T \hat{\mathbf{P}}_{j_{k+1}} \bar{\mathbf{x}}_{k+1} - \bar{\mathbf{x}}_k^T \hat{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_k \\ &\leq \bar{\mathbf{x}}_{k+1}^T \hat{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_{k+1} - \bar{\mathbf{x}}_k^T \hat{\mathbf{P}}_{j_k} \bar{\mathbf{x}}_k, \quad \hat{\mathbf{P}}_{j_k} \in \mathbb{P}_P \\ &= \bar{\mathbf{x}}_k^T [\mathbf{A}_{j_k}^T \mathbf{P}_{j_k} \mathbf{A}_{j_k} - \mathbf{P}_{j_k}] \bar{\mathbf{x}}_k \\ &\quad + 2\bar{\mathbf{x}}_k^T [\mathbf{A}_{j_k}^T \mathbf{P}_{j_k} \mathbf{B}_{j_k} + \mathbf{A}_{j_k}^T \hat{\mathbf{h}}_{j_k} - \hat{\mathbf{h}}_{j_k}] \\ &\quad + \mathbf{B}_{j_k}^T \mathbf{P}_{j_k} \mathbf{B}_{j_k} + 2\mathbf{B}_{j_k}^T \hat{\mathbf{h}}_{j_k} + d_{j_k} - d_{j_k} \\ &\leq -\bar{\mathbf{x}}_k^T \mathbf{W} \bar{\mathbf{x}}_k + C \leq -\lambda_{\min}(\mathbf{W}) \|\bar{\mathbf{x}}_k\|_2^2 + C \\ &= -\alpha_3 (\|\bar{\mathbf{x}}_k\|_2) + c_2. \end{aligned} \quad (26)$$

This shows that $V(\mathbf{x}_k)$ is a practical-Lyapunov function for the SAS with the switching scheme π_{MPC} for $\mathbb{X} = \mathbb{R}^n$ according to Definition 2. Therefore the SAS is globally UpAS according to Theorem 1. This completes the proof. ■

VI. SIMULATION RESULTS

A. Example 1: SAS With One Stable Subsystem

To illustrate the proposed MPC scheme SAS with 2 subsystems as follows is considered

$$\begin{aligned} \mathbf{A}_1 &= \begin{bmatrix} 0.5 & 0.2 \\ 0 & 0.8 \end{bmatrix}, \quad \mathbf{B}_1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \\ \mathbf{A}_2 &= \begin{bmatrix} 1 & 0.1 \\ -0.5 & 2 \end{bmatrix}, \quad \mathbf{B}_2 = \begin{bmatrix} -1.5 \\ -1 \end{bmatrix}. \end{aligned} \quad (27)$$

The spectral radius of the subsystems is 0.8 1.95 which indicates that only subsystem 1 is Schur stable. The weighting matrix is chosen as $\mathbf{Q}_{\sigma_k} = \mathbf{Q}_1$ where $\mathbf{Q}_1$ and $\mathbf{W}$ are computed using (20) for the parameter $C = 5$, and initial state is chosen as $\mathbf{x}_0 = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$. The simulation response of the proposed

MPC scheme and the enumeration-based MPC scheme for the prediction horizon $N = 5$ and $N = 10$ are given in Fig. 2(a) and 2(b).

We denote the time horizon or total simulation time by $N_T = 50$, and by $J = \sum_{k=0}^{N_T-1} \bar{\mathbf{x}}_{k|k}^T \mathbf{Q}_{\sigma_{k|k}} \bar{\mathbf{x}}_{k|k}$ the total cost over the simulation time which is used as a measure of overall performance. We denote by $\mu_E = |\mathbb{P}|$ and t_P, t_E the online computation time required (in seconds) for the proposed MPC and enumeration based MPC for a simulation over N_T . RC denotes the percentage reduction in computation time which is defined as $RC = \frac{t_E - t_P}{t_E} \times 100$. The simulations are performed in MATLAB 2022 on a 2.53 GHz Intel Core 2 Duo CPU. Table I shows the total cost J and the computation times t_P, t_E are averaged over 5 different initial conditions: $\begin{bmatrix} 10.9 \\ 5.1 \end{bmatrix}, \begin{bmatrix} 8.9 \\ 4.2 \end{bmatrix}, \begin{bmatrix} 11.4 \\ 5.3 \end{bmatrix}, \begin{bmatrix} 9.8 \\ 4.7 \end{bmatrix}, \begin{bmatrix} 10.8 \\ 4.7 \end{bmatrix}$ which

are randomly generated using $\mathbf{x}_0 = \begin{bmatrix} 10 \\ 5 \end{bmatrix} + \mathbf{r}(\mathbf{0}, \mathbf{I}_2)$ where $\mathbf{r}(\mathbf{0}, \mathbf{I}_2)$ is chosen as a 2-dimensional Gaussian random vector. Even though, the enumeration method always gives the best performance in terms of J for finite-horizon optimal control problems, when it comes to MPC this is not true. This can be observed from Table I since the proposed MPC scheme gives better performance for small prediction horizons compared to the enumeration-based MPC.

B. Example 2: Buck Converter

Here we consider a buck converter which is a DC-DC converter that can be modelled as a SAS with two subsystems

$$\dot{\mathbf{x}} = \mathbf{A}_{c_\sigma} \mathbf{x} + \mathbf{B}_{c_\sigma} \quad (28)$$

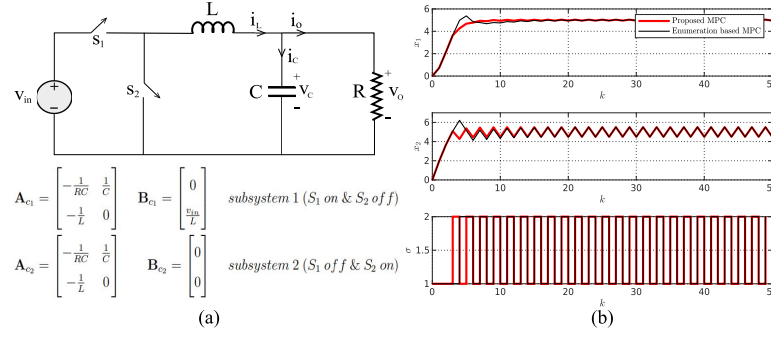


Fig. 3. (a) Buck converter circuit and subsystems (b) Response of buck converter with proposed MPC and enumeration-based MPC for $N = 10$.

where $t \in \mathbb{R}^+$ is the continuous time instant, $\mathbf{x} = \mathbf{x}(t) \in \mathbb{R}^2$ is the continuous state vector, $\sigma = \sigma(t) : \mathbb{R}^+ \rightarrow \mathbb{M} = \{1, 2\}$ is the continuous switching index that decides the active mode at time t , $\mathbf{A}_{c_v} \in \mathbb{R}^{n \times n}$, $\mathbf{B}_{c_v} \in \mathbb{R}^{n \times 1}$, $v \in \mathbb{M}$ are the system and input matrix for the subsystem v . Here the state vector $\mathbf{x} = \begin{bmatrix} v_C \\ i_L \end{bmatrix}$ for which the circuit diagram and subsystems are given in Fig. 3(a) (see [2] for a detailed derivation). The discrete-time SAS model of the buck converter can be obtained by discretizing (28) with a sampling time T which gives

$$\mathbf{x}_{k+1} = \mathbf{A}_{d_{\sigma_k}} \mathbf{x}_k + \mathbf{B}_{d_{\sigma_k}} \quad (29)$$

where $\mathbf{A}_{d_v} = e^{\mathbf{A}_{c_v} T}$ and $\mathbf{B}_{d_v} = \int_0^T e^{\mathbf{A}_{c_v} \lambda} \mathbf{B}_{c_v} d\lambda$, $v \in \mathbb{M}$. For a nonzero reference $\mathbf{x}_r \neq 0$, the error state is defined as $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_r$ and from (29) dynamics of the error state becomes

$$\mathbf{e}_{k+1} = \mathbf{A}_{\sigma_k} \mathbf{e}_k + \mathbf{B}_{\sigma_k} \quad (30)$$

where $\mathbf{A}_v = \mathbf{A}_{d_v}$ and $\mathbf{B}_v = \mathbf{B}_{d_v} + (\mathbf{A}_{d_v} - \mathbf{I}_n) \mathbf{x}_r$, $v \in \mathbb{M}$. By defining $\tilde{\mathbf{x}}_k = \begin{bmatrix} \mathbf{e}_k \\ 1 \end{bmatrix} \in \mathbb{R}^{n+1}$, we can transform the state equation (30) as in (2) which is then used for the MPC design.

For the buck converter, the parameters are chosen as $R = 1\Omega$, $C = 0.001F$, $L = 0.005H$, $V_{in} = 10V$, $x_{1r} = 5$ and $T = 1ms$. The simulation parameters are chosen as $\mathbf{x}_0 = [0 \ 0]^T$, $\mathbf{x}_r = [5 \ 5]^T$, and $\mathbf{Q}_2 = 3\mathbf{Q}_1$ where $\mathbf{Q}_1$ and $\mathbf{W}$ are computed using (20) for the parameter $C = 5$. The simulation response of the buck converter with the MPC schemes is given in Fig. 3(b) for $N = 10$.

VII. CONCLUSION

In this letter, a practically stabilizing MPC scheme is presented for discrete-time SASs which uses a pruning scheme with cost approximation to assure feasibility and stability. It is shown that the proposed pruning scheme results in globally UpAS dynamics for the SAS and the value function of the MPC scheme will be a practical Lyapunov function. Simulation results show that the MPC scheme results in lesser online computation with acceptable performance. Further work includes extending the MPC scheme by incorporating state constraints and obtaining less conservative bounds for stability.

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